Exact Phase, Projective, and Metric Atlases for a PT-Symmetric Dimer

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Abstract

For the autonomous balanced-gain/loss dimer $\mathrm{i}\dot{\psi} = H_{\gamma,\kappa}\psi$, $\kappa > 0$, we give an all-parameter dynamical classification. The scalar identity $H^2 = (\kappa^2 - \gamma^2)I$ yields the exact propagator, generic periods in the unbroken chamber, a rank-one nilpotent exceptional point, and the attracting/repelling eigenrays of the broken chamber.

1 Frozen system and exact propagator

The frozen release obstruction identifier is `HEN-0281`. Let

$$H = H_{\gamma,\kappa} = \begin{pmatrix} \mathrm{i}\gamma & \kappa \\ \kappa & -\mathrm{i}\gamma \end{pmatrix}, \qquad \kappa > 0, \quad \gamma \in \mathbb{R}, \qquad \mathrm{i}\dot{\psi} = H\psi.$$

Parity is $\mathcal{P} = \sigma_x$ and time reversal is componentwise complex conjugation, so $\mathcal{P}H^*\mathcal{P} = H$. Time is the physical propagation parameter; it is not adjusted to external data. Put $\delta = \kappa^2 - \gamma^2$.

Theorem 1 (Complete three-chamber atlas)**.** *The propagator is*

$$e^{-\mathrm{i}tH} = \begin{cases} \cos(\omega t)I - \mathrm{i}\omega^{-1}\sin(\omega t)H, & \delta > 0, \quad \omega = \sqrt{\delta}, \\ I - \mathrm{i}tH, & \delta = 0, \\ \cosh(\nu t)I - \mathrm{i}\nu^{-1}\sinh(\nu t)H, & \delta < 0, \quad \nu = \sqrt{-\delta}. \end{cases}$$

If $\delta > 0$, a ray having nonzero components in both eigenlines has least projective period π/ω, while its vector representative has least period $2\pi/\omega$. The eigenrays themselves are stationary. If $\delta = 0$, H is nonzero rank-one nilpotent with one eigenline, and every generalized state has linear leading growth. If $\delta < 0$, there are two fixed rays; one attracts every generic forward ray and the other repels it, with exponential rate 2ν in their affine ratio.

Proof. Multiplication gives

$$H^2 = (\kappa^2 - \gamma^2)I = \delta I.$$

The even and odd terms of the exponential series give the three formulae. For $\delta > 0$ the eigenvalues are $\pm\omega$. In their eigenbasis a generic component ratio is multiplied by $e^{2\mathrm{i}\omega t}$, proving the ray period; equality of both vector phases first occurs at $2\pi/\omega$. For $\delta = 0$, nonzero determinant-zero H has rank one and the exponential terminates. For $\delta < 0$ the eigenvalues are $\pm\mathrm{i}\nu$, so the evolution multipliers are $e^{\pm\nu t}$ and their ratio proves the final assertion. $\square$

This proof is uniform at the exceptional point: it does not diagonalize there or suppress the generalized eigenvector. It also prevents a common factor of two error between vector and ray periods.

References

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